考量螺槳動態旋轉之潛艇流場網格佈建策略研究

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摘　　要

潛艦螺槳推力與船體阻力計算之精確性能預測對其運動機動性與戰場隱蔽性之優劣評估至關重要。雖然目前計算流體力學數值方法已具備預測潛艦船體與螺槳交互作用水動力特性之能力，惟數值模擬結果的準確性，除取決於物理模型與數值計算方法外，亦與流場網格之佈建品質及解析能力密切相關。尤其在螺槳動態旋轉所產生之複雜非定常流場中，網格品質與配置策略更是影響數值計算精度、效能及結果可靠性之關鍵因素。本研究旨在探討考量螺槳動態旋轉效應之潛艇流場數值模擬網格佈建策略，以提升螺槳推進條件下潛艇周圍複雜流場之計算效率與數值模擬精度。網格解析能力及動態網格配置方式將直接影響螺槳推進性能、艇體阻力及尾流場計算結果，若潛艇與螺槳表面採用過度細緻之邊界層網格，雖可提升局部流場解析能力，卻將大幅增加網格數量與計算成本；反之，若前述物體周圍網格解析不足，則可能造成潛艇與螺槳流場無法獲得適當解析。本研究將以「Joubert BB2潛艇—Seiun Maru高歪斜螺槳」之整合構型為研究對象，建立適用於螺槳動態旋轉之計算域與網格佈建方法，針對艇體、螺槳及尾流區域建構細緻與粗糙等不同尺度之網格配置，並配合滑移網格（Sliding Mesh）等動態網格技術，模擬螺槳實際旋轉所造成之非定常流場變化，而藉由模擬結果與文獻資料之比較，研究成果將可建立一套兼顧流場解析能力與計算效率之潛艇螺槳動態旋轉網格佈建策略以作為後續潛艇推進性能分析、非定常流場研究及水下航行器螺槳—艇體流場數值模擬之基礎。有關研究成果概述，圖1為本研究所構建之兩種不同潛艇與螺槳網格配置比較說明，該圖左下部為非等向性細緻邊界層網格，而右下部則為網格尺寸較大之等向性非結構性網格。為先驗證數值方法之可靠性，本研究已分開執行螺槳與潛艇之個別流場模擬並與文獻資料(Kao)資料進行比較，在前述驗證算例中，潛艇航速為10.802 m/sec，螺槳轉速則為156.48 RPM，圖2為含螺槳與未含螺槳之流場模擬結果比較，兩者速度場分佈之主要差異出現在靠近螺槳船艉區域，另由圖3比較結果顯示，本論文之螺槳模擬結果與文獻資料相當接近(<5%)，而相較於等向性非結構性網格算例，細緻邊界層網格算例之模擬結果則與文獻資料較為相似；最後，圖4則展現潛艇在螺槳作動狀態下，利用前述兩種不同網格計算所得之潛艇阻力與螺槳推力之差異，由模擬結果得知，在前述模擬條件下，阻力與推力並不匹配，且兩者所得結果皆是呈現推力明顯大於阻力，若潛艇以特定航速航行，此項問題將可藉由動態調整螺槳轉速達成推阻匹配要求，如此將可深入探討潛艇與螺槳複雜交互作用流場問題，並藉此客觀評估網格佈建策略對計算效益之影響。

關鍵詞：計算流體力學、BB2潛艇、Seiun Maru高歪斜螺槳、交互作用流場

A Study on Grid Generation Strategies for Submarine Flow Field Simulations Considering Dynamic Propeller Rotation

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ABSTRACT

Accurate prediction of submarine propeller thrust and hull resistance is essential for evaluating submarine maneuverability and stealth performance. Although CFD methods can effectively predict submarine–propeller interaction hydrodynamics, the accuracy and efficiency of numerical simulations are strongly influenced by computational mesh quality and resolution, particularly for the complex unsteady flow generated by a rotating propeller. This study investigates mesh-generation strategies for submarine flow-field simulations considering dynamic propeller rotation. The Joubert BB2 submarine equipped with the Seiun Maru highly skewed propeller is adopted as the computational model. Two mesh configurations are investigated, including a finely resolved anisotropic boundary-layer mesh and a relatively coarse isotropic unstructured mesh. The sliding-mesh method is employed to simulate the actual propeller rotation and associated unsteady flow field. Figure 1 compares the two mesh configurations. To verify the numerical method, submarine and propeller simulations were performed

and compared with the published data of Kao. The submarine speed was 10.802 m/s and the propeller rotational speed was 156.48 rpm. Figure 2 compares the velocity fields with and without the propeller, showing significant differences mainly in the stern region near the propeller. As shown in Figure 3, the propeller results agree well with the published data, with an error of less than 5%. The finely resolved boundary-layer mesh also provides closer agreement than the isotropic unstructured mesh. Figure 4 compares the predicted hull resistance and propeller thrust for the two mesh configurations under propeller-operating conditions. The results show that the predicted thrust is substantially greater than the hull resistance. For a prescribed submarine speed, thrust–resistance matching can be achieved by adjusting the propeller rotational speed. The results demonstrate the importance of mesh configuration for accurately and efficiently predicting submarine–propeller interaction flow fields.

Keywords: CFD, BB2 Submarine, Seiun Maru Propeller, Interaction Flow Fields

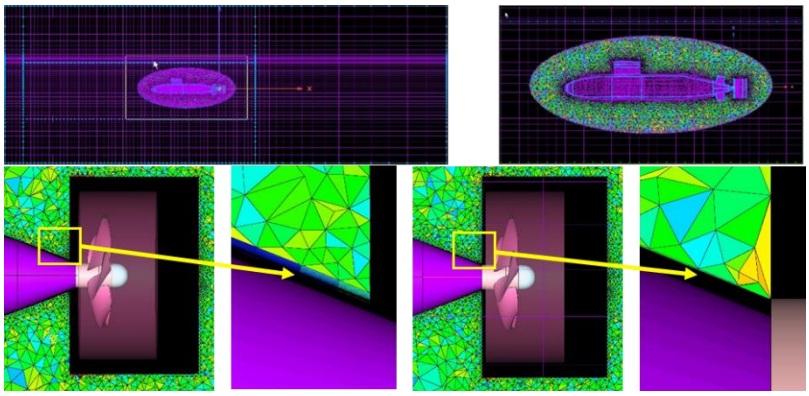


圖 1 潛艇與螺槳網格配置/非等向性細緻邊界層網格(左下圖)，等向性非結構性網格(右下圖)

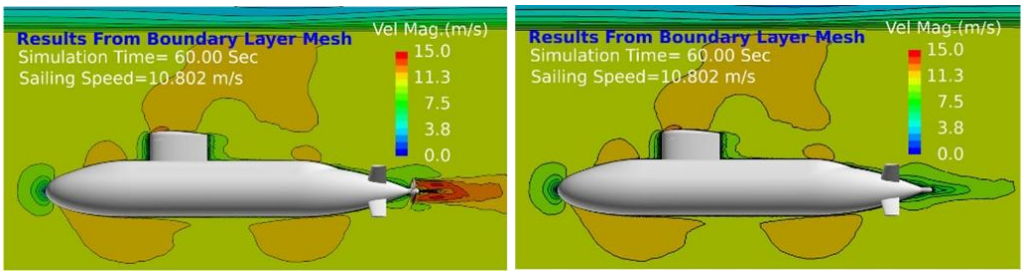


圖 2 潛艇流場比較/含螺槳運動流場(左圖)，未含螺槳運動流場(右圖)

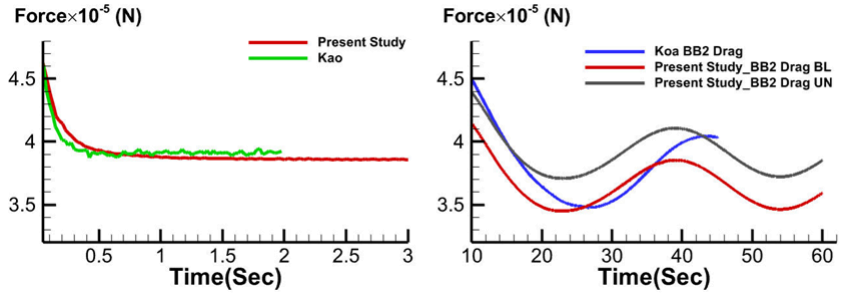


圖 3 螺槳流場模擬與文獻(Kao)結果比較(左圖)，潛艇流場模擬與文獻(Kao)結果比較(右圖)

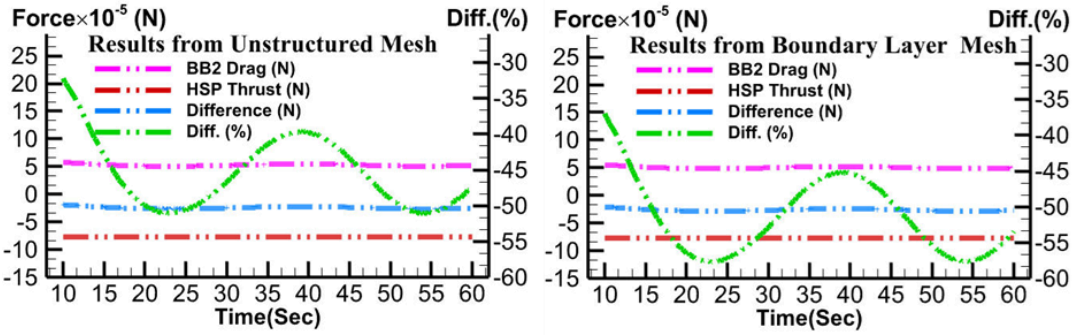


圖 4 潛艇阻力與螺槳推力差異/等向性非結構性網格(左圖)，非等向性細緻邊界層網格(右圖)